AI-Driven QoS Prediction in SD-WAN

Networks

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DECLARATION I, Radhika(Rishiv) Shitlani, hereby declare that this thesis, titled "AI-Driven QoS Prediction in SD-WAN Networks", and the work presented in it are entirely my own except where explicitly stated otherwise in the text, and that this work has not been previously submitted, in part or whole, to any university or institution for any degree, diploma, or other qualification.

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Abstract

Software-Defined Wide Area Networks (SD-WAN) are used to connect branch offices, cloud services, and data centres over different types of links. A key challenge in SD-WAN is Quality of Service (QoS). Important traffic such as voice or video needs low delay, low jitter, and low packet loss, while background traffic can normally tolerate worse performance. Many SD-WAN policies are still static or reactive. They act after a problem is already visible.

This thesis investigates whether machine learning can predict QoS behaviour in SD-WAN-like environments and support proactive QoS decisions. The work uses two public-data-based experiments. The first experiment uses the Zenodo 5G campus network dataset to predict achieved throughput from radio and traffic configuration features. This acts as a link-capacity prediction baseline. The second experiment uses a locally generated BNN-UPC/BNNetSimulator dataset to study per-class QoS delay for Gold, Silver, and Bronze traffic under different scheduling policies.

For the Zenodo dataset, Random Forest achieved the strongest throughput result among the tested models, with a cross-validated R^2 of approximately 0.94. XGBoost is close behind at 0.93. The dataset is useful for link capacity prediction, but it does not contain QoS class variation. For the BNN-UPC dataset, XGBoost acts as the strongest

explainable tree-based baseline, while the PyTorch feedforward neural network is the main neural-network path. Both models achieved similar performance on the log-transformed average delay target, with cross-validated R^2 values around 0.766–0.769.

The thesis also addresses a key evaluation issue: a single global metric can hide poor performance for one QoS class. Therefore, the BNN-UPC model is evaluated by QoS class, scheduling scenario, and scheduling policy. The results show that Gold and Silver delay are predicted much more accurately than Bronze delay, with per-class R^2 values of 0.914, 0.908, and 0.704 respectively on a lightly loaded dataset.

Beyond average delay, the evaluation is extended to two additional QoS targets: 90th-percentile delay (delay_p90) and jitter. delay_p90 shows moderate predictability for Gold and Silver ($R^2 \approx 0.64$–0.66), while jitter is found to be largely unpredictable from static network configuration features ($R^2 < 0.12$). This is a meaningful finding: jitter is driven by dynamic traffic burstiness that queue weights and scheduling policy alone cannot capture.

The Bronze SLA violation threshold is tuned from 100 ms to 60 ms based on a threshold sweep, improving Bronze Recall from 0.368 to 0.568 and F1 from 0.489 to 0.592. Packet loss evaluation is restricted to Bronze traffic, which is the only class that experiences meaningful loss in a realistic network (10.7% of Bronze flows in v1 have non-zero loss; Gold and Silver have near-zero loss). A binary classifier (predicting any loss versus no loss) achieves AUC-ROC of 0.916 for Bronze flows, confirming the model can discriminate loss-prone flows well even though exact loss-rate regression is not feasible from static features. Finally, a Layer 3 allocation recommender is implemented. It tests candidate WFQ bandwidth profiles and recommends the least-violating Gold/Silver/Bronze allocation under SLA thresholds. This connects QoS prediction to the original SD-WAN decision question: how much bandwidth should be allocated to each QoS class?

Contents

Chapter 1

Introduction

In today's digital world, businesses rely heavily on wide area networks (WANs) to connect their offices, data centers, and cloud services. Traditional WANs, which often use expensive and rigid MPLS links, are increasingly being replaced by Software-Defined Wide Area Networks (SD-WAN). SD-WAN offers more flexibility, cost savings, and centralized control. One of the biggest responsibilities of any WAN including SD-WAN is to ensure smooth and reliable network performance. This is referred to as Quality of Service (QoS), which includes important metrics like delay (latency), jitter, packet loss, and available bandwidth.

In networks, traffic patterns keep fluctuating since different applications have different needs, and the same network might suddenly be overloaded. For example, video conferencing needs minimal or zero packet loss along with low latency, while file downloads just need higher bandwidth. If this is not taken care of, user experience is impacted and organizations may face penalties for violating SLAs(Service Level Agreements). Therefore, maintaining good QoS is not only critical but challenging.

Currently, most SD-WAN solutions use static policies to configure QoS. This means that network policies are either set manually in advance or adjusted after a problem is detected or reported. The problem here is: by the time a QoS issue is noticed, it might already be too late and user experience is already impacted. This delay can affect real-time applications like voice or video calls, which need instant action. There's a clear need for a better, smarter solution.

This is where Artificial Intelligence (AI) and Machine Learning (ML) come in and can help to resolve the problem. These technologies can learn from historical data and detect patterns in the network traffic. If trained correctly, ML models can predict QoS problems before they happen. For example, if the system learns that high video traffic around 2 PM usually leads to congestion, it can automatically adjust bandwidth or reroute traffic ahead of time. This kind of prediction allows network operators to prevent problems instead of just reacting to them.

Scope and Context

This thesis explores how AI and ML techniques can be used to predict QoS in SD-WAN-like networks. The project began with a broad focus on QoS prediction, but the implemented work now has three connected layers. Layer 1 predicts link capacity using the Zenodo 5G dataset. Layer 2 predicts per-class delay using a BNN-UPC/BNNetSimulator dataset, using XGBoost as the strongest explainable tree-based baseline and PyTorch MLP as the main neural-network direction. Layer 3 uses the delay model to recommend a WFQ bandwidth allocation for Gold, Silver, and Bronze traffic.

Problem Statement

SD-WAN makes it easier to manage WANs, but maintaining QoS is still a challenge, especially during unexpected traffic surges or link failures. Most QoS-related decisions today are made either too late or based on static rules. Predicting QoS metrics in advance using AI could solve this problem but it's not clear which methods work best, how accurate they are, or how easy they are to implement in real networks.

Research Questions

This report aims to answer the following questions:

- **RQ1**: What types of AI or ML techniques are currently or can be used for predicting QoS in SD-WAN?
- **RQ2**: What are the benefits and limitations of these approaches in terms of prediction accuracy, scalability, and real-world use?
- **RQ3**: How can QoS prediction be effectively integrated into SD-WAN production environments to improve network performance?

Methodology

To explore these questions, I conducted a literature review using academic databases like IEEE Xplore, Scopus, and Google Scholar. I searched for terms such as "QoS prediction," "SD-WAN," "machine learning in networking," and "AI-based traffic forecasting." Papers published between 2020 and 2025 were prioritized, and older foundational works were included when relevant.

The implementation then followed an experimental approach. First, public datasets were reviewed and unsuitable proxy datasets were removed from the main modelling path. Second, supervised ML models were trained on the Zenodo and BNN-UPC datasets, including both classical ML models and a PyTorch MLP. Third, the BNN-UPC results were evaluated using QoS-aware metrics, including per-class error, policy-specific slicing, RMSLE, and SLA violation precision. Finally, a simple allocation recommender was built to connect prediction to an

SD-WAN control decision. A planned next step is to run the same allocation evaluation using the MLP and compare its recommendations with XGBoost.

Structure of the Document

This document is structured as follows:

- Chapter 1 introduces the research topic, its scope, the problem statement, and the key research questions.
- Chapter 2 provides background information on SD-WAN, QoS metrics, and the role of AI in modern network management.
- Chapter 3 presents the literature review, including an analysis of academic work on QoS prediction in SD-WAN using AI and ML techniques.
- Chapter 4 outlines the methodology, including dataset selection, feature engineering, model design, QoS-aware evaluation, and the allocation recommender.
- Chapter 5 presents the experimental setup and results for the Zenodo and BNN-UPC datasets.
- Chapter 6 details the project plan, timeline, tools, and supervision checkpoints.
- Chapter 7 discusses the risks and limitations of the project, including data availability, model complexity, and real-world constraints.
- Chapter 8 concludes the report by summarizing findings, highlighting real-world impact, and identifying directions for future work.

Chapter 2

Background

This chapter provides foundational knowledge needed to understand the scope and significance of this thesis. It covers the key concepts of Software-Defined Wide Area Networks (SD-WAN), Quality of Service (QoS) metrics in networking, and the role of Artificial Intelligence (AI) in network management. These concepts establish the context for the research discussed in later chapters.

2.1 Software-Defined Wide Area Networks (SD-WAN)

SD-WAN is an approach to wide area networking that uses software-based control to manage network traffic over multiple connection types, including MPLS, broadband, LTE, and 5G. Unlike traditional WANs that rely on expensive private circuits, SD-WAN provides centralized orchestration and greater flexibility, enabling cost-effective routing based on application priorities and real-time conditions.

A typical SD-WAN architecture includes:

- **Edge devices:** Deployed at branch locations to route traffic intelligently.
- **Central controller:** Makes policy-based routing decisions and pushes configurations to edge devices.
- **Overlay and underlay networks:** Logical tunnels (overlay) established on top of physical connections (underlay).

SD-WAN improves application performance by enabling dynamic path selection, fast failover, traffic shaping, and visibility across all network segments.

2.2 Quality of Service (QoS)

QoS in networking refers to the ability to provide different priority levels to different applications, users, or data flows, and to guarantee a certain level of performance for critical traffic. It is crucial for maintaining service-level agreements (SLAs), especially in modern, cloud-centric environments.

Common QoS metrics include:

- **Latency:** Time taken for a packet to travel from source to destination.
- **Jitter:** Variation in packet delay across a network.
- **Packet loss:** Percentage of packets lost during transmission.
- **Bandwidth:** Maximum data transfer rate of a link.

Effective QoS management ensures that delay-sensitive applications like VoIP and video conferencing are not negatively affected by congestion or fluctuating traffic.

2.3 Artificial Intelligence in Network Management

Artificial Intelligence (AI), especially Machine Learning (ML), is increasingly being used in networking to automate and optimize operations. AI techniques can analyze large volumes of network telemetry data to detect anomalies, classify traffic, and predict future network states.

In the context of SD-WAN, AI is being used to:

- Predict traffic trends and adjust policies proactively.
- Recommend optimal routing paths based on current and expected conditions.
- Forecast QoS metrics and warn of potential SLA violations.

AI models used for QoS prediction typically include supervised learning algorithms (like decision trees and neural networks) and reinforcement learning (used in adaptive routing and load balancing). These models require training on historical network data to make accurate forecasts.

2.4 Importance of QoS Prediction in SD-WAN

Modern enterprises rely on applications that require consistent and reliable network performance. Static or reactive QoS mechanisms are often too slow to prevent performance issues. QoS prediction enables a more proactive approach—by forecasting issues before they occur, it becomes possible to take preventive actions like re-routing traffic or scaling resources.

This capability is especially important in SD-WAN environments where network conditions vary across different links and access types. Accurate QoS prediction can help optimize resource usage, reduce costs, and improve user experience.

Chapter 3

Literature Review

This chapter presents a detailed review of the existing research related to the use of Artificial Intelligence (AI) and Machine Learning (ML) techniques for predicting Quality of Service (QoS) in Software-Defined Wide Area Networks (SD-WAN). The aim is to understand what work has already been done, what methods and models have been proposed, and where gaps still exist. This will help assess the current state of the field and how it supports or challenges the research questions introduced earlier.

3.1 Search Strategy and Selection Process

To gather the literature for this review, a structured search was carried out using academic databases such as IEEE Xplore, Scopus, and Google Scholar. The following search terms were used:

- "QoS prediction" AND "SD-WAN"
- "AI for QoS" AND "Software-defined networking"
- "Machine learning traffic classification" AND "network performance"

The search focused on papers published between 2018 and 2025 to ensure up-to-date relevance. Articles before 2018 were excluded unless they provided foundational knowledge. From an initial pool of documents, key research papers were selected based on the following criteria:

- Peer-reviewed publications
- Clear focus on QoS, SD-WAN, or software-defined networking
- Use of AI or ML for prediction, classification, or optimization
- Availability of experimental results, models, or architecture designs

Although the topic of QoS prediction using AI in SD-WAN is promising, the literature review revealed a limited number of high-quality research papers that directly focus on this specific intersection. Most existing studies either concentrate on general SDN environments or address QoS management without the use of AI or ML techniques. Nonetheless, this review provides valuable insights and highlights key research gaps that warrant further investigation.

3.2 Overview of Selected Papers

The selected papers cover a range of methods and applications—some focus on predicting traffic load, others on routing or load balancing. Table 3.1 categorizes these papers by their core approach and relevance to the research questions.

3.3 AI and ML Methods for QoS Prediction (RQ1)

Many studies have applied AI techniques to network traffic prediction and management. For instance, Rahman et al. (2019) [6] used Random Forest and deep learning models to predict how many virtual network functions (VNFs) are needed during a traffic surge. Their models achieved over 95% accuracy and could prevent service degradation by scaling resources in advance.

Similarly, Zheng et al. (2022) [7] developed a deep neural network to classify traffic flows based on their latency sensitivity. They then routed this traffic using a QoS-Aware Routing algorithm that selected paths depending on the required delay tolerance. This combination of classification and path assignment showed how AI can directly improve QoS outcomes.

These studies clearly support RQ1 by showing that supervised learning and deep learning can be applied to anticipate traffic needs and assign optimal resources in advance.

3.4 Suitability and Real-World Limitations (RQ2)

Although the above techniques work well in controlled settings, there are challenges when using them in real networks. For example, the deep reinforcement learning (DRL) model by Dinh et al. (2024) [2] well in simulations and maintains safe traffic routing using control barrier functions (CBFs). However, DRL requires a lot of training time and GPU support, which may not be available in all SD-WAN deployments.

In contrast, Quang et al. (2023) [4] proposed SABE which is a lightweight method that uses queuing theory to estimate how much bandwidth is left on a link without sending test traffic. This approach avoids complex AI models but may be less flexible when traffic patterns become unpredictable.

This contrast shows that while AI models are powerful, their complexity and resource requirements must be carefully considered before deployment, which directly addresses RQ2.

3.5 Integration into SD-WAN Systems (RQ3)

Several papers also discussed how these models can be integrated into actual SD-WAN systems. For instance, Rahman et al. (2019) [6] simulated an SD-WAN scenario where their ML models helped reduce both QoS violations and cloud leasing costs by scaling VNFs only when necessary.

Quang et al. (2022) [3] presented a practical example of an enterprise moving from traditional MPLS to SD-WAN. Their central controller dynamically adjusted both routing and QoS scheduling based on the current network status. This shows that AI-based prediction and decision-making can be realistically applied in SD-WAN setups to improve efficiency and reliability.

These findings support RQ3, confirming that AI is not only a research tool but can also be integrated into live systems if the environment supports it.

3.6 Research Gaps and Observations

Despite the progress seen in these papers, several gaps were noticed:

- Most research focuses on predicting a single QoS metric like latency or bandwidth, rather than multiple metrics at once.
- Most results are based on simulations. Real-world deployment results in live SD-WAN systems are rare.
- Only few research documents are focusing on using Artificial Intelligence in the prediction or correction of QOS.
- Public datasets related to SD-WAN traffic are hard to find, limiting reproducibility.

- Few works compare the AI predictions directly against decisions made by standard SD-WAN controllers.

Filling these gaps could make QoS prediction models more robust, more generalizable, and more useful for industry.

3.7 Summary Of Literature Review So Far

This review found that AI and ML models especially supervised learning, deep learning, and reinforcement learning, are promising tools for predicting QoS trends in SD-WAN. The models help forecast traffic load, classify application needs, and adjust routing or resource allocation dynamically. However, practical deployment still faces challenges due to training complexity, hardware requirements, and lack of real-world validation.

Although the number of direct studies focused on AI-based QoS prediction in SD-WAN remains limited, the relevance of the topic continues to grow as networks become more software-defined and application-aware. Recent work such as MTEAL [1] also shows that state-of-the-art SD-WAN traffic engineering is moving beyond simple machine learning models towards graph neural networks, reinforcement learning, and multi-dimensional QoS optimisation. Based on the review and in consultation with Dr. Jawad Manzoor, this project will keep simple models as baselines only, while exploring stronger non-linear and deep-learning-inspired approaches where feasible. The scope will be carefully managed so that the chosen datasets, techniques, and evaluation metrics remain realistic within the resources and time frame of a master's-level project.

Table 3.1: Summary and Categorisation of Selected Literature

Paper	**AI/ML Technique**	**Main Focus**	**Related RQ(s)**
Rahman et al. [6]	Random Forest, DNN	Predicting VNF requirements to maintain QoS and reduce cost	RQ1, RQ2, RQ3
Zheng et al. [7]	Deep Neural Network + ILP	Classifying application types and assigning QoS-based routes	RQ1, RQ2
Dinh et al. [2]	DRL (PPO, DDPG) with CBF	Safe traffic routing using AI that respects network constraints	RQ1, RQ2
Quang et al. [4]	SABE (Queuing theory-based estimation)	Estimating available bandwidth using passive measurements	RQ2, RQ3
Quang et al. [3]	Centralised Optimization + QoS scheduler	Managing QoS priorities and routing using SD-WAN controller	RQ2, RQ3
Chen et al. [1]	GNN + MARL	SD-WAN traffic engineering with multi-dimensional QoS metrics	RQ1, RQ2, RQ3

Chapter 4

Methodology

This chapter describes the method used in this project. The aim is not to build a commercial SD-WAN product. The aim is to test whether machine learning can predict QoS outcomes and whether those predictions can support a simple SD-WAN bandwidth-allocation decision.

The project uses a layered methodology:

- **Layer 1: Link capacity prediction.** The Zenodo 5G dataset is used to predict achieved throughput from radio and traffic configuration features.
- **Layer 2: Per-class delay prediction.** A BNN-UPC/BNNetSimulator dataset is used to predict delay for Gold, Silver, and Bronze traffic classes under different scheduling policies.
- **Layer 3: Allocation recommendation.** The trained delay model is used as a surrogate model to test candidate WFQ weight profiles and recommend a Gold/Silver/Bronze bandwidth allocation.

4.1 Dataset Selection

A real enterprise SD-WAN dataset was not available. Therefore, public datasets and simulation-based datasets were used.

4.1.1 Zenodo 5G Dataset

The Zenodo 5G campus network dataset (record 13754300) was used as the first dataset. It contains real 5G throughput measurements from two testbeds. It includes offered throughput, achieved throughput, jitter, packet loss, radio bandwidth, slot configuration, uplink/downlink direction, and one-way-delay packet measurements.

This dataset is useful for predicting link capacity. However, it has an important limitation for this thesis: all rows belong to the same application type. It does not contain Gold, Silver, and Bronze QoS class variation. Therefore, it cannot answer the per-class QoS allocation question by itself.

4.1.2 BNN-UPC / BNNetSimulator Dataset

The second dataset was generated locally using the official BNNetSimulator Docker image. This was necessary because the official BNN-UPC Challenge download links were not available. BNNetSimulator is the same type of simulator used to generate the BNN-UPC network modelling datasets, so it is suitable for a reproducible simulation-based QoS experiment.

The generated BNN-UPC dataset contains 29,280 per-flow records. Each row represents one source-destination flow. The dataset includes three QoS classes:

- **Gold:** ToS 0, highest priority.
- **Silver:** ToS 1, medium priority.

- **Bronze:** ToS 2, lowest priority.

Four scheduling scenarios were generated:

- Fixed WFQ weights.
- Random WFQ profiles.
- Mixed SP/WFQ/DRR policies with skewed ToS distribution.
- Mixed SP/WFQ/DRR policies with equal ToS distribution.

This dataset is referred to as the **v1 dataset** and is used for the main QoS scheduling and allocation experiments under a light-to-moderate load regime.

4.1.3 Datasets Not Used as Main Inputs

CICIDS2017 was explored earlier but was dropped from the main modelling path. It is an intrusion-detection dataset, not a QoS dataset. Using it would require too much reverse engineering of QoS labels. MAWI was also considered, but it does not contain direct Gold/Silver/Bronze labels and would require substantial extra feature engineering. Therefore, the final implemented work focuses on Zenodo and BNN-UPC.

4.2 Feature Engineering

For the Zenodo dataset, the main target is `actual_throughput_mbps`. The input features include radio configuration, offered throughput, direction, testbed metadata, latency, packet count, and flow duration. Leakage columns such as `recommended_bandwidth_percent`, `jitter_ms`, and `packet_loss_percent` are removed when predicting throughput.

For the BNN-UPC dataset, the main target is `log_avg_delay`. The log transform is used because raw `avg_delay` is heavy-tailed. Bronze traffic can experience much larger delay spikes than Gold and Silver traffic, especially under congestion.

The BNN-UPC feature set used for modelling contains:

- `tos` and `qos_class`
- `offered_bandwidth`
- `time_distribution`
- `routing_hops`
- `n_nodes`
- `max_avg_lambda`
- `link_bandwidth`
- `scheduling_policy`
- `tos_queue_weight`
- `min_tos_weight`

Outcome columns such as `avg_delay`, `jitter`, `packet_loss_rate`, delay percentiles, and `actual_bandwidth` are removed from the input features to avoid leakage.

4.3 Models

The following models were trained and compared:

- **DummyRegressor:** A naive baseline that predicts the training mean.

- **Linear Regression:** A simple interpretable baseline.
- **Support Vector Regression (SVR):** A kernel-based non-linear model.
- **Random Forest Regressor:** A tree ensemble model.
- **XGBoost Regressor:** A gradient-boosted tree model.
- **PyTorch MLP:** A feedforward neural network with hidden layers of 128, 64, and 32 neurons.

The simple ML models are important because they are easier to explain and are suitable for tabular data. XGBoost is treated as the strongest explainable tree-based baseline. The PyTorch MLP is treated as the main neural-network path for the project, because it tests whether a feedforward neural model can learn the same QoS delay behaviour and later support per-class allocation decisions.

4.4 Evaluation Methodology

The first evaluations used common regression metrics: MAE, RMSE, R^2, cross-validation scores, and inference time. However, for BNN-UPC this is not enough. Gold, Silver, and Bronze traffic have different priorities and different delay behaviour. A single global metric can hide poor performance for one class.

Therefore, the BNN-UPC evaluation was extended using the following methods:

- **Per-class MAE:** Errors are reported separately for Gold, Silver, and Bronze flows.
- **Policy-specific slicing:** Results are sliced by scheduling scenario and scheduling policy.

- **RMSLE:** Root Mean Squared Logarithmic Error is reported because delay has a heavy-tailed distribution.
- **SLA violation precision:** Continuous delay predictions are converted into binary SLA violation triggers. Precision measures how trustworthy the trigger is when the model predicts a violation.

This evaluation better matches the SD-WAN use case. It shows not only whether the model is accurate overall, but also whether it is reliable for high-priority traffic.

4.4.1 Multi-Metric QoS Evaluation

Average delay is the primary QoS target, but real SD-WAN environments also care about tail delay and jitter. Therefore, the evaluation is extended to two additional targets:

- **delay_p90:** The 90th percentile of packet delay. It represents the worst-case experience for 10% of packets, which is more relevant than mean delay for real-time applications.
- **jitter:** Variation in packet delay. High jitter degrades voice and video quality even when mean delay is acceptable.

Both targets are heavy-tailed like average delay, so a log transformation is applied before training and predictions are back-transformed for reporting. MAE, RMSE, RMSLE, and R^2 are reported per QoS class for each target. This allows direct comparison of how predictable each metric is from the available network configuration features.

4.5 Layer 3 Allocation Recommender

The final implemented component is a simple allocation recommender. It is model-agnostic: it can use a trained BNN-UPC delay model, such as XGBoost or the PyTorch MLP, to test candidate WFQ profiles such as:

- 60/30/10
- 50/40/10
- 33/33/34
- 25/65/10
- 80/15/5

Each profile represents a possible Gold/Silver/Bronze bandwidth allocation. For each profile, the recommender rewrites the queue-weight features, predicts class-level delay, checks SLA thresholds, and ranks the profiles. The current reported allocation results use the trained delay model already evaluated in this thesis. The next planned version will run the same allocation workflow with the PyTorch MLP and compare the per-class recommendations against XGBoost. This directly addresses the original SD-WAN question: how much bandwidth should be allocated to each QoS class?

Chapter 5

Experiments and Results

This chapter presents the experiments completed so far. The results are organised by the three project layers: Zenodo throughput prediction, BNN-UPC per-class delay prediction, and Layer 3 QoS allocation recommendation.

5.1 Layer 1: Zenodo Throughput Prediction

5.1.1 Dataset and Setup

The Zenodo 5G campus network dataset [5] was used to predict achieved throughput. The processed dataset contains 604 rows after uplink and downlink measurements are split into separate records. The target is `actual_throughput_mbps`.

The dataset was split into 80% training and 20% testing using a fixed random seed. Five-fold cross-validation was also used on the training split. Leakage columns such as `recommended_bandwidth_percent`, `jitter_ms`, and `packet_loss_percent` were dropped. Constant metadata columns such as `protocol`, `application_type`, `link_type`, and `sdr` were also removed because they do not vary across the dataset.

5.1.2 Results

Table 5.1 shows the Zenodo throughput results after dropping constant metadata features.

Table 5.1: Zenodo Throughput Prediction Results (Target: `actual_throughput_mbps`)

Model	**MAE**	**RMSE**	R^2	**CV** R^2
DummyRegressor	36.90	43.88	−0.007	−0.011
Linear Regression	10.55	14.15	0.895	0.851
SVR (RBF)	5.27	8.16	0.965	0.925
Random Forest	2.76	6.13	0.980	0.936
XGBoost	3.20	6.32	0.979	0.934

The Zenodo results show that supervised ML can predict achieved throughput well from radio and traffic configuration features. Random Forest gives the best held-out R^2, while XGBoost is close. However, this dataset contains only one application type and no Gold/Silver/Bronze class variation. Therefore, it is useful as a link-capacity prediction experiment, but it cannot answer the per-class QoS allocation question on its own.

5.2 Layer 2: BNN-UPC Delay Prediction

5.2.1 Dataset and Setup

The BNN-UPC dataset was generated using BNNetSimulator. It contains 29,280 per-flow records. Each flow has a QoS class label: Gold, Silver, or Bronze. The target used for modelling is `log_avg_delay`, because raw delay is heavy-tailed.

The feature set includes offered bandwidth, ToS, QoS class, routing hops, topology size, traffic distribution, scheduling policy, and queue weights. Outcome columns such as raw delay, jitter, loss rate, delay percentiles, and actual bandwidth were removed from model inputs.

5.2.2 Global Model Results

Table 5.2 shows the global BNN-UPC results on `log_avg_delay`.

Table 5.2: BNN-UPC Global Results (Target: `log_avg_delay`)

Model	**MAE**	**RMSE**	R^2	**CV** R^2
Linear Regression	0.227	0.415	0.685	0.690
SVR (RBF)	0.174	0.391	0.721	0.721
Random Forest	0.181	0.364	0.758	0.764
XGBoost	0.185	0.359	0.765	0.766
PyTorch MLP	0.185	0.357	0.766	0.769

The PyTorch MLP is slightly ahead of XGBoost on the global score, but the difference is small. This means that the neural-network path is competitive, while XGBoost remains an important explainable tree-based baseline for this tabular dataset. The useful comparison is therefore not only which model has the highest global score, but also whether the MLP and XGBoost behave similarly for each QoS class.

5.3 QoS-Aware Evaluation

A single global metric is not enough for BNN-UPC. Bronze delay is much more volatile than Gold and Silver delay. If all rows are pooled together, Bronze errors can dominate the score and hide how the model behaves for high-priority traffic.

Therefore, the XGBoost model was evaluated using out-of-fold predictions and sliced by QoS class, scheduling scenario, and scheduling policy. This provides the current explainable baseline for class-aware evaluation. The same class-aware evaluation was then applied to the PyTorch MLP so that the neural model is assessed per class, not only through a global score.

5.3.1 Per-Class Results

Table 5.3 shows per-class results. The metric values are on the delay scale in milliseconds, except R^2, which is computed on the log-delay target.

Table 5.3: BNN-UPC Per-Class Evaluation using XGBoost

QoS Class	Rows	MAE Delay	RMSLE	R^2 Log Delay
Gold	4,592	3.05 ms	0.160	0.914
Silver	8,991	3.48 ms	0.168	0.908
Bronze	15,697	20.68 ms	0.451	0.704

This result is important. Gold and Silver are predicted accurately, but Bronze is much harder. This confirms that global model scores alone are not enough for QoS evaluation.

5.3.2 Policy and Scenario Results

Table 5.4 shows that the mixed-policy scenario is the hardest to predict.

Table 5.4: BNN-UPC Scenario-Specific Evaluation using XGBoost

Scenario	**Rows**	**MAE Delay**	R^2 **Log Delay**
A_wfq_fixed	6,560	6.16 ms	0.796
B_wfq_profiles	8,240	12.01 ms	0.758
C_mixed_policy	7,600	21.32 ms	0.743
D_mixed_equal	6,880	9.95 ms	0.779

Table 5.5 shows results sliced by scheduling policy. Strict Priority (SP) achieves the best R^2 because SP flows follow a deterministic priority order that the model can learn well. WFQ is harder because queue weights vary across profiles and interact with traffic load in more complex ways.

Table 5.5: BNN-UPC Scheduling Policy Evaluation using XGBoost (v1 Dataset)

Policy	**Rows**	**MAE Delay**	**RMSLE**	R^2 **Log Delay**
SP (Strict Priority)	8,120	8.27 ms	0.282	0.806
DRR (Deficit Round Robin)	1,840	8.01 ms	0.302	0.678
WFQ (Weighted Fair Queueing)	19,320	14.90 ms	0.378	0.756

5.3.3 SLA Violation Precision

For SD-WAN use, exact delay prediction is not the only concern. The model should also identify when a flow is likely to violate an SLA. The continuous delay prediction was converted into a binary trigger using these thresholds: Gold > 30 ms, Silver > 50 ms, and Bronze > 60 ms.

The Bronze threshold was tuned from an initial value of 100 ms to 60 ms following a sensitivity sweep. Table 5.6 summarises how precision, recall, and F1 change as the Bronze threshold is lowered. Lowering the threshold increases the number of flows treated as violations in the trigger evaluation, making the alert more sensitive and improving recall. The 60 ms value was chosen because it keeps Bronze strictly more lenient than Silver (50 ms) while significantly improving recall.

Table 5.6: Bronze SLA Threshold Sensitivity (Gold=30 ms, Silver=50 ms fixed)

Bronze Threshold	**Precision**	**Recall**	**F1**
100 ms	0.726	0.368	0.488
70 ms	0.614	0.476	0.536
60 ms	0.619	0.568	0.592
50 ms	0.582	0.735	0.650

Table 5.7 shows the final SLA violation trigger results using the tuned thresholds.

Table 5.7: SLA Violation Trigger Results using XGBoost (Tuned Thresholds)

QoS Class	**Threshold**	**Precision**	**Recall**	**F1**
Gold	30 ms	0.994	0.899	0.944
Silver	50 ms	0.823	0.639	0.720
Bronze	60 ms	0.619	0.568	0.592

Gold precision remains very high (0.994), meaning the model is correct 99.4% of the time when it predicts a Gold SLA violation. This is critical for SD-WAN policy triggers, because false positives for Gold could cause unnecessary rerouting of high-priority traffic. Bronze recall improves from 0.368 (at 100 ms) to 0.568 (at 60 ms), a 55% improvement, while precision drops only modestly from 0.726 to 0.619. The threshold adjustment makes the alerting system more sensitive for Bronze, which is appropriate given that Bronze is best-effort traffic and missing a violation is more acceptable than for Gold.

5.3.4 Model Comparison: XGBoost vs PyTorch MLP

The results above use XGBoost as the strongest explainable tree-based baseline. To evaluate the neural-network direction of the project, the same QoS-aware analysis was repeated for a PyTorch feedforward MLP (three hidden layers of 128, 64, and 32 units, ReLU activations, dropout 0.1, AdamW optimiser, early stopping). The outer out-of-fold evaluation protocol and metric functions are identical to the XGBoost evaluation; the MLP additionally uses an internal validation split inside each training fold for early stopping. This makes the per-class numbers directly comparable while avoiding validation leakage.

Table 5.8 compares per-class delay prediction.

Table 5.8: Per-Class Delay Prediction: XGBoost vs PyTorch MLP (v1 Dataset, 5-fold out-of-fold)

QoS Class	**Model**	**MAE (ms)**	**RMSLE**	R^2
Gold	XGBoost	3.05	0.160	0.914
Gold	MLP	3.08	0.161	0.912
Silver	XGBoost	3.48	0.168	0.908
Silver	MLP	3.42	0.169	0.908
Bronze	XGBoost	20.68	0.451	0.704
Bronze	MLP	20.60	0.450	0.705

On delay prediction the two models are practically very similar: they agree to within roughly 0.1 ms of MAE and 0.002 of R^2 on every class. Neither modelling family has a meaningful advantage in this experiment. This is consistent with the well-documented finding that gradient-boosted trees and neural networks can perform comparably on small, static, tabular feature sets such as this one.

Table 5.9 compares SLA violation detection at the tuned thresholds.

Table 5.9: SLA Violation Detection: XGBoost vs PyTorch MLP (Tuned Thresholds)

QoS Class	Threshold	Model	Precision	Recall	F1
Gold	30 ms	XGBoost	0.994	0.899	0.944
Gold	30 ms	MLP	0.979	0.908	0.942
Silver	50 ms	XGBoost	0.823	0.639	0.720
Silver	50 ms	MLP	0.791	0.715	0.751
Bronze	60 ms	XGBoost	0.619	0.568	0.592
Bronze	60 ms	MLP	0.591	0.608	0.599

On SLA detection a consistent pattern emerges: the MLP trades a small amount of precision for higher recall, most visibly on the harder Silver and Bronze classes (Silver recall rises from 0.639 to 0.715; Bronze recall from 0.568 to 0.608). In other words, the neural model raises slightly more alerts and therefore catches more true violations at the cost of more false alarms. Which behaviour is preferable depends on the operational priority: XGBoost's near-perfect Gold precision (0.994) is better when false positives on high-priority traffic must be avoided, while the MLP's higher recall is preferable when missing a violation is the greater concern.

Overall, XGBoost is retained as the recommended model because it matches the MLP's accuracy while offering interpretability (feature importance) and substantially lower training cost. The MLP comparison confirms that the strong per-class results are a property of the data and features, not an artefact of a single model family.

5.4 Multi-Metric QoS Evaluation (v1 Dataset)

Beyond average delay, the BNN-UPC v1 dataset was used to evaluate delay_p90 and jitter per QoS class. The same out-of-fold XGBoost pipeline was used for both targets, with a log transform applied before training.

5.4.1 delay_p90 Results

Table 5.10 shows the per-class results for 90th-percentile delay prediction.

Table 5.10: Per-Class delay_p90 Prediction Results using XGBoost (v1 Dataset)

QoS Class	**Rows**	**MAE (ms)**	**RMSLE**	R^2
Gold	4,592	6.64 ms	0.205	0.635
Silver	8,991	7.29 ms	0.215	0.657
Bronze	15,697	38.22 ms	0.499	0.258

Gold and Silver show moderate predictability ($R^2 \approx 0.64$–0.66), which is lower than for average delay. This is expected because the 90th percentile captures rarer, more variable congestion events. Bronze tail delay is much harder to predict ($R^2 = 0.258$), following the same pattern as average delay but amplified.

5.4.2 Jitter Results

Table 5.11 shows the per-class results for jitter prediction.

Table 5.11: Per-Class Jitter Prediction Results using XGBoost (v1 Dataset)

QoS Class	**Rows**	**MAE (ms)**	**RMSLE**	R^2
Gold	4,592	0.14 ms	0.176	0.119
Silver	8,991	0.16 ms	0.190	0.084
Bronze	15,697	3.11 ms	0.678	0.014

Jitter R^2 is very low across all classes (0.01–0.12). This is a meaningful finding rather than a modelling failure. The BNN-UPC features describe static network configuration: queue weights, scheduling policy, and offered bandwidth. These features determine mean delay well but do not capture the moment-to-moment traffic burstiness that drives jitter. Jitter prediction would require dynamic traffic features such as instantaneous queue occupancy or inter-arrival time statistics, which are not present in this dataset. This is discussed further as a limitation and future work direction.

5.5 Bronze Packet Loss Analysis

In the v1 dataset, packet loss is realistic: Gold and Silver flows experience near-zero loss (4 and 93 loss events respectively out of 13,583 combined rows), while Bronze flows experience non-zero loss in 1,684 out of 15,697 rows (10.7%). This matches the expected behaviour of a network where high-priority traffic (Gold, Silver) is protected and only best-effort Bronze traffic suffers under load.

Following supervisor guidance, packet loss evaluation is restricted to Bronze traffic only. Gold and Silver loss events are too rare and too small to train a reliable model on, and artificially forcing loss onto high-priority classes would not reflect a realistic network scenario.

5.5.1 Regression Attempt

Regression on `packet_loss_rate` for Bronze flows produced $R^2 = -0.130$. This is a meaningful finding: packet loss events in this dataset are driven by rare, bursty buffer overflow events. The static network configuration features (queue weights, scheduling policy, offered bandwidth) determine mean delay well but cannot capture when a particular burst will exceed the buffer. Predicting the exact loss rate requires dynamic features such as instantaneous buffer occupancy or inter-arrival time statistics, which are not present in the dataset.

5.5.2 Binary Loss Classification for Bronze

A more appropriate framing is binary classification: predicting whether a Bronze flow will experience any packet loss at all (loss > 0). This converts the problem from estimating an unpredictable rate to a detection task. The class balance is 89.3% no-loss versus 10.7% loss, so class weighting (`scale_pos_weight` $= 8.3$) is applied.

Table 5.12 shows 5-fold stratified cross-validation results for XGBoost and Random Forest.

Table 5.12: Bronze Binary Loss Classification Results (v1 Dataset, 5-fold CV)

Model	**Precision**	**Recall**	**F1**	**AUC-ROC**
XGBoost (default threshold)	0.777	0.328	0.461	0.914
XGBoost (class-weighted)	0.342	0.865	0.490	0.916
Random Forest (balanced)	0.663	0.389	0.490	0.895

The AUC-ROC of 0.916 for class-weighted XGBoost confirms the model has strong discriminative ability: it can rank Bronze flows by loss probability effectively. The low recall at the default 0.5 decision threshold (0.328) is a threshold choice, not a model failure. Using class weighting raises recall to 0.865 at the cost of lower precision (0.342), which may be appropriate for an alerting system where missing a loss event is more harmful than a false alarm.

This shows that while exact loss-rate regression is not feasible from static features, the model can reliably identify which Bronze flows are at elevated risk of experiencing packet loss.

5.6 Layer 3: Allocation Recommender

The final experiment uses the trained BNN-UPC delay model as a surrogate model for allocation decisions. The recommender is designed so that it can use either an explainable model such as XGBoost or a neural model such as the PyTorch MLP. The current reported allocation results use the trained delay model already evaluated in this chapter. Several WFQ profiles were tested: 60/30/10, 50/40/10, 33/33/34, 25/65/10, and 80/15/5. Each profile was applied to the traffic pattern, and the model predicted the expected delay for Gold, Silver, and Bronze.

The default SLA thresholds for this allocation experiment were Gold < 20 ms, Silver < 50 ms, and Bronze < 100 ms. This recommender run keeps the original allocation thresholds; rerunning the recommender with the tuned Bronze threshold of 60 ms is left as a follow-up comparison.

Table 5.13: Layer 3 WFQ Allocation Recommendation Results

Rank	Profile	Feasible	Gold Mean	Silver Mean	Bronze Mean
1	60/30/10	No	24.47 ms	24.22 ms	30.81 ms
2	80/15/5	No	24.99 ms	30.62 ms	31.27 ms
3	25/65/10	No	25.02 ms	24.28 ms	30.81 ms
4	33/33/34	No	25.12 ms	24.55 ms	33.92 ms
5	50/40/10	No	25.27 ms	24.80 ms	30.81 ms

No tested profile fully satisfies the strict Gold threshold of 20 ms. The recommender therefore selects 60/30/10 as the least-violating profile. This is still useful because the system can show both a recommendation and an infeasibility warning. In practice, this could tell an SD-WAN controller that changing WFQ weights alone may not be enough and that rerouting or extra capacity may be needed. A future comparison should run the same profile search using the PyTorch MLP and check whether the neural model recommends the same allocation.

5.7 Summary of Findings

The completed experiments show seven main findings:

- Zenodo is useful for link capacity prediction, but not for per-class QoS allocation.
- BNN-UPC provides a useful per-class QoS scheduling dataset. Global metrics hide class-level differences; Bronze is much harder to predict than Gold and Silver.
- QoS-aware evaluation with per-class MAE, RMSLE, and SLA violation precision is essential for reporting honest model behaviour for SD-WAN.
- delay_p90 is moderately predictable ($R^2 \approx 0.64$–0.66 for Gold/Silver), but jitter is not predictable from static configuration features ($R^2 < 0.12$). This is a feature gap, not a modelling failure.
- Bronze SLA recall improves significantly when the violation threshold is tuned from 100 ms to 60 ms (Recall: $0.368 \rightarrow 0.568$, F1: $0.489 \rightarrow 0.592$). The 60 ms threshold was selected to keep Bronze strictly more lenient than Silver (50 ms) while providing a meaningful recall improvement.
- Packet loss is only meaningful for Bronze traffic in a realistic network. Regression on packet_loss_rate is not feasible from static features ($R^2 = -0.130$), but binary classification (loss vs. no loss) achieves AUC-ROC =

0.916 for Bronze flows, confirming the model can identify loss-prone flows reliably.

- The PyTorch MLP matches XGBoost per class (delay R^2 within 0.002 on every class) but does not beat it, while trading a little precision for higher recall on the harder Silver and Bronze classes. XGBoost is retained as the recommended model for its equal accuracy, interpretability, and lower training cost.

Chapter 6

Project Plan and Timeline

This chapter outlines the structured plan and timeline for completing the capstone project. The schedule is aligned to the final submission deadline of 31st August 2026 and includes time for dataset validation, model development, evaluation, documentation, and supervisor feedback.

6.1 Project Milestones

The table below presents the proposed timeline of key project phases and deliverables:

Table 6.1: Timeline and Key Deliverables

Time Period	Status	Activities and Deliverables
June 2025	Done	Finalized project idea and submitted project proposal
July–Nov 2025	Done	Background research, literature review, and initial dataset exploration
Dec 2025–Jan 2026	Done	Finalized project scope, created GitHub repository, reviewed candidate datasets
Feb–Mar 2026	Done	Implemented preprocessing pipeline and baseline modelling scripts
Apr–May 2026	Done	Processed the Zenodo dataset, generated the BNN-UPC dataset using BNNetSimulator, trained classical ML models and a PyTorch MLP, added QoS-aware evaluation, and built the Layer 3 allocation recommender
June 2026	Planned	Refine the evaluation, compare MLP and XGBoost per-class behaviour, review the Layer 3 recommender results, and incorporate supervisor feedback
July 2026	Planned	Complete the full thesis draft, improve figures and tables, and decide whether any small additional experiment is needed
August 2026	Planned	Final proofreading, submit thesis by 31st August 2026, and prepare for viva

6.2 Tools and Resources

The following tools and platforms will be used throughout the project:

- **Programming:** Python (pandas, scikit-learn, XGBoost, PyTorch)
- **Environment:** JupyterLab / VS Code
- **Data Analysis:** matplotlib, seaborn
- **Documentation:** Overleaf for LaTeX-based report writing
- **Version Control:** GitHub (for code management and backups)

6.3 Supervision and Checkpoints

Regular meetings and email check-ins with the supervisor, Dr. Jawad Manzoor, will be planned around key milestones. Ad-hoc consultation via email or MS Teams will be arranged when feedback is needed between reviews.

Chapter 7

Risks and Limitations

This chapter outlines the potential risks and limitations associated with the proposed project, along with mitigation strategies. Identifying these factors in advance helps to set realistic expectations and provides contingency planning to ensure the project stays on track.

7.1 Data Availability

One of the primary risks is the limited availability of real-world SD-WAN traffic datasets, particularly those with labelled QoS metrics. While public datasets provide valuable traffic data, they may not directly reflect enterprise SD-WAN-specific behaviour or QoS classifications.

An internal workplace dataset could not be obtained, so this project will rely on public datasets. This limitation will be explicitly discussed in the thesis to avoid over-claiming real-world SD-WAN generalisability.

Mitigation: The project uses two public sources. The Zenodo 5G dataset is used for link capacity prediction. BNN-UPC is used for QoS scheduling and per-class delay prediction. The official BNN-UPC download links were unavailable, so the dataset was generated locally using the official BNNetSimulator environment and the same output format. CICIDS2017 was dropped because it is an intrusion detection dataset. MAWI is kept as future work because it would require extra labelling and would expand the scope too much.

7.2 Model Complexity and Training Resources

Advanced models like deep neural networks or ensemble methods may require significant computational power, especially for hyperparameter tuning and training on larger datasets.

Mitigation: The project uses simple and explainable baselines first, including Linear Regression, SVR, Random Forest, and XGBoost. XGBoost is kept as the strongest explainable tree-based baseline. The PyTorch MLP is the main neural-network direction for future development, especially for comparing per-class allocation decisions. More complex transformer or reinforcement learning models are not treated as the main contribution.

7.3 Feature Engineering Challenges

Not all features listed in the methodology may be available or relevant across all datasets. Inconsistent data schemas, missing values, or incompatible units may limit the effectiveness of certain features.

Mitigation: Separate preprocessing pipelines are used for each dataset. Zenodo features are focused on signal quality and throughput. BNN-UPC features are focused on traffic load, routing hops, ToS class, scheduling policy, and queue weight. This avoids forcing different datasets into one artificial schema.

7.4 Evaluation in Real-World Context

Since the proposed work is likely to use offline datasets, the final results may not directly translate to real-time SD-WAN deployment without further validation. Simulations might not capture dynamic, production-grade behaviors such as live failovers or multi-path routing decisions.

Mitigation: The project focuses on a reproducible proof of concept using offline data. The evaluation includes class-aware metrics, scenario-specific slicing, RMSLE, and SLA violation precision. These metrics are closer to operational SD-WAN use than a single global regression score, but live controller validation remains future work.

7.5 Time Constraints

Any delays in data access, tool setup, or model convergence could impact the final delivery timeline.

Mitigation: A flexible and modular project plan has been developed, with early-stage tasks front-loaded to reduce later risk. Regular checkpoints with the supervisor will help track progress and address blockers quickly.

Chapter 8

Conclusion

This thesis investigated the use of machine learning models to predict Quality of Service (QoS) outcomes in SD-WAN-like network environments. The final project has three layers: link capacity prediction, per-class delay prediction, and allocation recommendation.

The literature review identified a broad range of AI and ML techniques applied to QoS prediction and SD-WAN traffic engineering, ranging from classical supervised learning to graph neural networks and reinforcement learning. The review confirmed that while deep learning and transformer-based models represent the current state of the art, interpretable baseline models remain essential as comparison benchmarks and are computationally feasible for near-real-time deployment.

The first dataset is the Zenodo 5G campus network dataset. It is useful for predicting link capacity from radio and traffic features. On this dataset, tree-based and kernel models perform strongly, with Random Forest and XGBoost achieving test R^2 values around 0.98. This shows that the measured link conditions contain a strong signal for throughput prediction.

The second dataset is the BNN-UPC dataset generated using BNNetSimulator. This dataset is more important for the original research question because it includes Gold, Silver, and Bronze traffic classes under different scheduling policies and queue weights. On log-transformed average delay, XGBoost and the PyTorch MLP both achieve strong global results, with cross-validation R^2 around 0.77. XGBoost is useful because it is easier to explain, while the MLP is important because it represents the neural-network direction of the project.

The QoS-aware evaluation is a key part of the work. It shows that a single global score is not enough. Gold and Silver delay are predicted with low error, while Bronze is more difficult because its delay is more volatile and heavy-tailed. SLA violation precision is especially strong for Gold traffic, which is useful because Gold flows are normally the most important class for SD-WAN policy decisions.

The Layer 3 allocation recommender uses the trained BNN-UPC delay model to compare candidate WFQ weight profiles. In the tested traffic pattern, none of the profiles fully satisfy the strict Gold SLA threshold of 20 ms. The recommender therefore selects the least-violating profile and reports that weight adjustment alone may not be enough. This connects the predictive model back to the original research question: how much bandwidth should be allocated to each QoS class? The current reported allocation results should be extended by running the same recommender with the PyTorch MLP and comparing the per-class allocation decisions against XGBoost.

8.1 Real-World Impact or Benefit

The core value of this project lies in enabling proactive Quality of Service (QoS) management in SD-WAN environments through AI-driven predictions. In current networks, QoS violations are often addressed reactively, after user experience is already affected. This project aims to change that by predicting issues in advance using real-time flow-level features.

Practical Benefits in Real Networks:

- **Proactive Network Management:** Helps operators act before congestion or SLA violations occur.
- **Cost Optimization:** Enables more informed bandwidth allocation and may reduce unnecessary overprovisioning.
- **Improved QoE (Quality of Experience):** Real-time predictions allow better support for critical applications like video calls, VoIP, and live collaboration tools.
- **Scalability:** The approach is suitable for large enterprise SD-WANs that span multiple locations and require dynamic QoS adjustments.

8.2 Future Work

Based on the current findings, future research should explore the following directions:

- **Bronze Loss Classifier Deployment:** The current Bronze binary loss classifier achieves AUC-ROC = 0.916. The next step is to select the optimal decision threshold (e.g. using precision-recall curve analysis) and integrate the classifier alongside the delay model so that the SD-WAN controller receives both a delay prediction and a loss-risk score for each Bronze flow.
- **Neural Allocation Recommender:** Extend the Layer 3 recommender to use the PyTorch MLP, then compare per-class allocation results against XGBoost using Gold/Silver/Bronze SLA thresholds.

- **Better Bronze Delay Prediction:** Bronze delay is harder to predict than Gold and Silver ($R^2 = 0.704$ vs. 0.914 and 0.908). A dedicated model with class-weighted training or a separate Bronze-only model may reduce the Bronze MAE further. The tuned SLA threshold (60 ms) already improves recall, but better per-class models remain worth exploring.
- **Jitter Prediction with Dynamic Features:** Jitter is not predictable from static configuration features. Future work should explore adding dynamic traffic features such as instantaneous queue depth or inter-arrival time statistics.
- **More Allocation Profiles:** Test a wider search space of WFQ weights and include rerouting or extra-capacity decisions when no feasible weight profile exists.
- **Real-Time Integration:** Integrate the predictive model into an SD-WAN controller or policy engine.
- **Cross-Domain Generalisation:** Test the approach on more topologies, real traffic traces, or additional public datasets such as MAWI.

By addressing these improvements, the project could evolve into a plug-in or microservice within SD-WAN platforms that provides predictive QoS as a built-in capability.

8.3 Final Remarks

AI and ML hold significant promise for improving QoS in SD-WAN by enabling proactive, predictive management rather than reactive troubleshooting. While the current body of research is still emerging and somewhat fragmented, there is a strong foundation to build upon. With more focused work and collaboration between academia and industry, AI-driven QoS prediction can become a core feature of future network systems.

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